VAIBHAV KULKARNI AI Engineer

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EXPERIENCE

Cloud Security Web - Associate AI Software Engineer — October 2025 - Present

- Engineered a production-grade **cross-platform Electron.js application (Windows/macOS)** for real-time medical speech transcription and EHR integration, used by [X]+ clinicians across [X] facilities.
- Integrated **NVIDIA Parakeet TDT 0.6B-v2 INT8 ASR, pyannote, and 3D-Speaker** via sherpa-onnx, enabling **CPU-only ONNX inference** for production workloads with [X]% lower latency and no GPU dependency.
- Built a real-time **speaker diarization pipeline** covering ML inference, speaker tracking, result processing, and live visualization, achieving [X]% speaker-attribution accuracy.
- Architected and deployed a scalable AI inference platform using Azure Container Apps and Service Bus, enabling on-demand transcription workers, distributed processing, fault isolation, and scaling to [X]+ concurrent sessions while reducing infrastructure cost by [X]%.
- Designed secure, low-latency AI application architecture using AudioWorklet, sandboxed Electron IPC, AES-256-GCM, and safeStorage, implementing secure data handling and protection controls for sensitive clinical data.
- Implemented **OpenTelemetry observability** and maintained **80–90% automated test coverage** across unit, integration, performance, and accessibility testing.

Trubot Electronics - AI Software Engineer — February 2024 – February 2026

- Designed and built a production **ERP/POS platform from scratch**, owning frontend, backend, database architecture, authentication, business workflows, AI integration, reporting, and deployment.
- Architected a multi-provider LLM platform using self-hosted Ollama (Qwen/Nemotron) and Google Gemini, implementing provider abstraction, model routing, automatic failover, retries, and configurable inference workflows, sustaining [X]% uptime across [X]+ daily AI requests.
- Built **multimodal document intelligence** for extracting structured inventory data from PDFs, images, and XLSX files, automating bulk inventory creation and cutting manual data entry time by [X]%.
- Developed core **ERP/POS workflows** spanning repair management, inventory, sales, returns, invoicing, customer/supplier management, attendance, payroll, and financial reporting for [X]+ staff across [X] store locations.
- Designed a **unified inventory architecture** across retail and repair workflows and implemented **JWT authentication with RBAC** for Owner, Receptionist, Incharge, and Technician roles.
- Built financial aggregation and document automation pipelines for **revenue, COGS, gross-profit reporting, invoices, receipts, and email delivery**, using **React, FastAPI, REST APIs, SQLite, and Node.js**.

PROJECTS

ElizaOS Agent - AI Career Counsellor [**GitHub Link**] — September 2025

- Designed and implemented a task-oriented AI agent using ElizaOS, integrating prompts, tools, conversational memory, and multi-step task orchestration for personalized career guidance.
- Applied agent orchestration patterns aligned with Azure AI Foundry, handling [X]+ multi-step career-planning scenarios end to end.

PrepWise-AI - Full-Stack Project (MERN) [**Live Link**] — August 2025

- Built an **AI-driven conversational system** using **Google Gemini LLMs** for dynamic interview Q&A generation, serving [X]+ users.
- Implemented **prompt engineering** and JSON structured outputs behind secure backend APIs, improving response consistency by [X]%.
- Deployed production-grade AI services with authentication, logging, and monitoring.

TECHNICAL SKILLS

AI Engineering & Generative AI: LLM application development, AI agents & agent orchestration, MCP, RAG, prompt engineering, structured outputs, function/tool calling, contextual memory, semantic search, vector embeddings, document intelligence, multimodal AI, model evaluation, AI observability, inference optimization
AI/ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, LangGraph, LlamaIndex, ElizaOS, MLflow
Cloud & MLOps: Microsoft Azure, Azure AI Foundry, Azure OpenAI, Azure Container Apps, Azure Service Bus, Azure Kubernetes Service, Azure Application Insights, Docker, CI/CD, Apache Airflow, ETL pipelines, data pipelines, model deployment, model monitoring, GCP, Agent development kit (ADK)
Backend & APIs: Python, FastAPI, Node.js, Express.js, REST APIs, JSON, API development, API integrations, enterprise workflows
Data & Databases: Pandas, NumPy, PySpark, PostgreSQL, MongoDB, Redis, Supabase, Azure Cosmos DB, vector databases
Programming Languages: Python, JavaScript, TypeScript, C, C++, CUDA
Development & Engineering: Git, GitHub, Agile/Scrum, SDLC, testing, debugging, monitoring, production deployment & maintenance

COMMUNITY & CONTRIBUTIONS

- Hacktoberfest 2024 Contributor:
  Merged 4+ pull requests in MERN open-source projects, earned Hacktoberfest badges.
- Tech Community Engagement:
  Active participant in **GDG Hyderabad** and **Google Build with AI** events.
- Mentorship:
  Conducted coding and software engineering workshops for 50+ junior developers, providing technical guidance on application development, debugging, and engineering practices.

EDUCATION

Bachelor of Engineering - Information Technology (CGPA: 6.18) June 2018 - May 2022
Maturi Venkata Subba Rao College,
Osmania University, Hyderabad, Telangana